Principles of Uncertainty

Homework 1 & 2

Timon Bubnič (63220035) · October 25, 2026

Problem 2.2 (Textbook, ch 2)

Solution. Let $\mathcal{C}$ be a collection of subsets of Ω. Then $\mathcal{C} = \sigma(\mathcal{C})$ if and only if $\mathcal{C}$ is a σ-algebra.

($\Rightarrow$) $\sigma(\mathcal{C})$ is a σ-algebra by definition.

($\Leftarrow$) We always have $\mathcal{C} \subseteq \sigma(\mathcal{C})$. Since $\mathcal{C}$ is a σ-algebra containing $\mathcal{C}$, and $\sigma(\mathcal{C})$ is the smallest such, $\sigma(\mathcal{C}) \subseteq \mathcal{C}$. □

Problem 2.3 (Textbook, ch 2)

Solution. Suppose $C \subseteq D$. Let

$$\chi = \{\text{all } \sigma\text{-algebras containing } C\}, \qquad \delta = \{\text{all } \sigma\text{-algebras containing } D\}.$$

Then $\delta \subseteq \chi$: every σ-algebra that contains D also contains C, since $C \subseteq D$ (the reverse doesn't hold in general). An intersection over a larger family is smaller, so

$$\sigma(C) = \bigcap \chi \subseteq \bigcap \delta = \sigma(D).$$

□

Problem 2.5 (Textbook, ch 2)

Solution. **a)** Let $\Omega = [0, 1]$ and let $\mathcal{F}_3$ consist of all countable subsets of Ω and all subsets of Ω with a countable complement.

(1) $\emptyset \in \mathcal{F}_3$ because it is countable.

(2) Let $A \in \mathcal{F}_3$.
(i) A is countable $\implies$ $(A^c)^c = A$ is countable $\implies$ $A^c \in \mathcal{F}_3$.
(ii) A is not countable. Since $A \in \mathcal{F}_3$, it has a countable complement, so A^c is countable $\implies$ $A^c \in \mathcal{F}_3$.

(3) Let $\{A_i\}_{i=1}^{\infty}$ be a countable sequence of elements of $\mathcal{F}_3$:
(i) All A_i are countable. $\implies$ $\bigcup_{i=1}^{\infty} A_i$ is also countable so it is $\in \mathcal{F}_3$.

(ii) There exists $k \in \{1, 2 \dots\}$ such that A_k is uncountable, therefore A_k^c must be countable. By De Morgan law we have:

$$(\bigcup_{i=1}^{\infty} A_i)^c = \bigcap_{i=1}^{\infty} A_i^c \subseteq A_k^c.$$

Since A_k^c is countable its subset must also be countable. Hence $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}_3$. Because $\mathcal{F}_3$ satisfies all 3 conditions it is a σ-algebra.

b) The tuple $(\Omega, \mathcal{F}_3)$ is a measurable space by part (a), since $\mathcal{F}_3$ is a σ-algebra on Ω. We have to check that P is a probability measure on it: If we take $\emptyset \in \mathcal{F}_3$, it is countable, therefore $P(\emptyset) = 0$. If we take $\Omega \in \mathcal{F}_3$ it isn't countable, however $\Omega^c = \emptyset$ is countable so $P(\Omega) = 1 - P(\Omega^c) = 1$. What is left is to check countable additivity.

Countable additivity. Let $A_1, A_2, \cdots \in \mathcal{F}_3$ be pairwise disjoint and $A = \bigcup_i A_i \in \mathcal{F}_3$.
(i) All A_i are countable. Then A is countable, so $P(A) = 0 = \sum_i P(A_i)$.
(ii) Some A_j is not countable, hence it's complement A_j^c is, so $P(A_j) = 1, P(A_j^c) = 0$. For $i \neq j$ disjointness gives $A_i \subseteq A_j^c$, which is countable, so $P(A_i) = 0$. Thus $\sum_i P(A_i) = 1$. Moreover $A_j \subseteq A$ gives $A^c \subseteq A_j^c$, so A^c is countable and $P(A) = 1$. Hence $P(A) = \sum_i P(A_i)$.

Therefore $(\Omega, \mathcal{F}_3, P)$ is a probability space. □